Lab 5-Evidence Analysis & Chain of Custody

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Tools: Velociraptor, FTK Imager | Date: 02 April 2026

1. Velociraptor — Live DFIR Platform:

What is Velociraptor?

Velociraptor is an open-source Digital Forensics and Incident Response (DFIR) tool originally developed by
Mike Cohen and maintained by Rapid7. It provides a powerful server-client architecture that allows security
analysts to perform remote forensic collection, live system interrogation, and threat hunting across large fleets of endpoints simultaneously. Communication is encrypted over TLS, and all queries are executed using VQL
(Velociraptor Query Language) — a SQL-like language purpose-built for forensic artifacts.

Key Features of Velociraptor:

FEATURE	DESCRIPTION
Live Network Analysis	Real-time netstat queries showing all TCP/UDP connections with PID correlation
Process Forensics	Full process tree with executable paths, command lines, and parent PIDs
File System Analysis	Search, hash, and collect files across endpoints using glob patterns
Registry Examination	Read and monitor Windows registry keys for persistence mechanisms
Event Log Collection	Collect and parse Windows Event Logs (Security, System, PowerShell)
Memory Forensics	Collect VAD memory regions and PE files loaded in memory
Bulk Hunting	Run VQL hunts across thousands of endpoints simultaneously
Timeline Analysis	Correlate timestamps across multiple artifact sources

2. FTK Imager — Forensic Acquisition Platform

What is FTK Imager?

FTK Imager (Forensic Toolkit Imager) by Exterro (formerly AccessData) is the industry-standard forensic evidence acquisition tool. It creates bit-for-bit forensic images of storage media while maintaining evidence integrity through cryptographic hashing (MD5, SHA1, SHA256). FTK Imager supports multiple image formats and provides both GUI and CLI interfaces for flexibility in forensic workflows

Key Features of FTK Imager

FEATURE	DESCRIPTION
Disk Imaging	Physical & logical drive imaging with E01/DD/AFF/SMART format support
Memory Capture	Live RAM acquisition including pagefile — preserves volatile evidence
Hash Verification	MD5, SHA1, SHA256 computed during imaging and verified post-acquisition

Image Mounting	Mount forensic images as read-only virtual drives for examination
Preview Mode	Browse file system, registry, and MFT without full imaging
Export	Export individual files, folders, or directory listings from images
CLI Support	Command-line interface for scripted and automated acquisitions

2.2 Lab Environment Topology:

HOST / ROLE	IP ADDRESS	OS	PURPOSE
Analyst Workstation	10.0.2.1	Ubuntu 22.04 / Kali	Velociraptor Server + Analyst machine
Server-Z (Target VM)	10.0.2.20	Windows Server 2019	Compromised endpoint under investigation
VirtualBox / VMware	—	Hypervisor on Host	Virtualization platform for the lab

a). Velociraptor Network Connection Analysis

Download Velociraptor Binary (Linux Analyst Machine)

```
# ■■■ On Linux Analyst Machine ■■■
```

```
# ■■■ Start the Velociraptor server ■■■
```

2.3 EVIDENCE ANALYSIS — Velociraptor Network Investigation

All VQL queries below are executed on Velociraptor Server GUI targeting the client SERVER-Z (IP: 10.0.2.20).
Path: GUI
→ Clients → SERVER-Z → New Collection → Generic.Client.VQL

#	ACTION	WHERE / DETAIL
1	Login	Open browser → https://10.0.2.1:8889 → Login with admin credentials
2	Select Client	Left panel → 'Clients' → Search '10.0.2.20' or 'SERVER-Z' → Click client
3	New Collection	Top tabs → 'Collected Artifacts' → Click green '+' button
4	Select Artifact	Search 'Generic.Client.VQL' → Select it → Paste VQL in 'Query' field
5	Launch	Click 'Launch' button — collection starts immediately
6	View Results	Click the completed collection → 'Results' tab → View or export data

Query 1 Full Netstat Dump
Purpose: Retrieve ALL network connections from the Windows VM (10.0.2.20).
VQL — Basic Netstat

```
-- ■■■ VQL Query 1: All Network Connections
■■■
```

```
Raddr.IP AS RemoteIP,
Raddr.Port AS RemotePort,
Status
FROM netstat
WHERE Status = 'ESTABLISHED'
AND Raddr.IP != '0.0.0.0'
AND Raddr.IP != '127.0.0.1'
AND Raddr.Port NOT IN (80, 443, 53, 22, 445, 135, 3389, 8080, 8443)
```

VQL Query 4 — Process & Network Full Correlation
This is the most important query — it joins process data with network data to reveal exactly which executable is
responsible for each network connection, including its full path and the user context it runs under.
VQL — Process + Network Correlation (Most Important)

```
-- ■■■ VQL Query 4: Full Process + Network Join ■■■■■■■■■■■■■■■■■■■■■■■■■■■
-- Correlates pslist() with netstat to show process + connection together
SELECT p.Pid AS PID,
p.Name AS ProcessName,
p.Exe AS ExecutablePath,
p.Username AS RunningAs,
p.CommandLine AS CommandLine,
n.Laddr.IP AS LocalIP,
n.Laddr.Port AS LocalPort,
n.Raddr.IP AS RemoteIP,
n.Raddr.Port AS RemotePort,
n.Status AS ConnectionStatus
FROM pslist() AS p
JOIN netstat AS n ON p.Pid = n.Pid
WHERE n.Raddr.IP != '0.0.0.0'
AND n.Status = 'ESTABLISHED'
ORDER BY p.Pid
```

Query 5 C2 Beacon & Malware Port Detection
This query hunts specifically for connections on ports commonly used by Command-and-Control (C2) frameworks such as Metasploit, Cobalt Strike, Empire, and custom malware.

```
-- ■■■ VQL Query 5: Common C2 / Malware Port Detection ■■■■■■■■■■■■■■■■■■■■
SELECT Pid,
Name,
Raddr.IP AS RemoteIP,
Raddr.Port AS RemotePort,
Status,
timestamp(epoch=CreateTime) AS ConnectionTime
FROM netstat


WHERE Raddr.Port IN (
4444, -- Metasploit default reverse shell
1234, -- Common backdoor port
31337, -- 'Elite' - historical trojan port
6666, -- IRC bots / C2
7777, -- Common RAT port
8888, -- Common C2 alternate
1337, -- Leet speak - custom malware
```

```
9999, -- Remote access tools
443, -- Include HTTPS for TLS tunneled C2
4443 -- Cobalt Strike HTTPS listener
)
OR (Status = 'ESTABLISHED'
AND Raddr.IP =~ '^(?!10\\.|172\\.1[6-9]\\.|192\\.168\\.)')
```

Query 6 Processes Running from Suspicious Locations
VQL — Suspicious Process Path Detection

```
-- ■■■ VQL Query 6: Malware often runs from Temp/AppData/Downloads ■■■■■■■■
SELECT p.Pid,
p.Name,
p.Exe,
p.CommandLine,

p.Username,
n.Raddr.IP AS RemoteIP,
n.Raddr.Port AS RemotePort,
n.Status
FROM pslist() AS p
JOIN netstat AS n ON p.Pid = n.Pid
WHERE p.Exe =~ '(?i)(\\temp\\|\\tmp\\|\\appdata\\|
\\downloads\\|\\public\\|\\users\\public)'
AND n.Raddr.IP != '0.0.0.0'
```

Suspicious Connection Indicators — Analysis Guide:

INDICATOR	SEVERITY	DESCRIPTION
Uncommon Port Numbers	HIGH	Connections on 4444, 1234, 8080, 9999, 31337 — typical C2 framework defaults
Unknown Remote IPs	HIGH	External IPs not matching known business services or CDN ranges
System Process Outbound	CRITICAL	svchost.exe, lsass.exe, csrss.exe making direct internet connections
PowerShell Connections	HIGH	powershell.exe with outbound ESTABLISHED connections — possible living-off-the-land
Temp/AppData Process	CRITICAL	Any process running from %TEMP%, %APPDATA% making network connections
Non-standard HTTPS	MEDIUM	HTTPS on non-443 port — possible C2 traffic hiding in encrypted channel
Regular Beacon Interval	HIGH	Same remote IP with connections every 30/60/300 seconds — C2 beaconing
Unusual Process Parent	MEDIUM	Process spawned by wrong parent (e.g., Word.exe → cmd.exe → powershell.exe)

Exporting Evidence Results from Velociraptor:

#	METHOD	PROCEDURE

1	GUI Export	Collection → Results tab → 'Download CSV' button → Save as NetworkLog_SERVER-Z_2026-03-30.csv
2	JSON Export	Collection → Results tab → 'Download JSON' button → For structured data processing
3	VQL to File	Use write_csv() VQL function to save directly on the endpoint (see code below)
4	Hash File	Compute SHA256 of exported file immediately after saving (see PowerShell below)

VQL — Export to CSV on Endpoint

```
-- ■■■ VQL: Export Results Directly to File on Endpoint ■■■■■■■■■■■■■■■■■■■
LET results <= SELECT Pid, Name,
Laddr.IP AS LocalIP, Laddr.Port AS LocalPort,
Raddr.IP AS RemoteIP, Raddr.Port AS RemotePort,
Status
FROM netstat WHERE Status = 'ESTABLISHED'
SELECT * FROM write_csv(
filename='C:/Forensics/Evidence/netstat_SERVER-Z_2026-03-30.csv',
query={ SELECT * FROM results }
)
```

POWERSHELL — SHA256 Hash Evidence File

```
# ■■■ PowerShell: Compute SHA256 of exported evidence file ■■■■■■■■■■■■■■■■■
$filePath = 'C:\Forensics\Evidence\netstat_SERVER-Z_2026-03-30.csv'
$sha256 = Get-FileHash -Path $filePath -Algorithm SHA256
$md5 = Get-FileHash -Path $filePath -Algorithm MD5
Write-Host '==============================================='
Write-Host 'EVIDENCE FILE HASH VERIFICATION'
Write-Host '==============================================='
Write-Host "File : $($sha256.Path)"
Write-Host "SHA256 : $($sha256.Hash)"

Write-Host "MD5 : $($md5.Hash)"
Write-Host "Date : $(Get-Date -Format 'yyyy-MM-dd HH:mm:ss UTC')"
Write-Host '==============================================='
# Save hash to sidecar file
"$($sha256.Hash) $($sha256.Path)" | Out-File -FilePath "$filePath.sha256"
```

2.4 EVIDENCE ACQUISITION — FTK Imager:

Create Forensic Disk Image of SERVER-Z (10.0.2.20)

1. Launch FTK Imager as Administrator
   - Start Menu → All Programs → AccessData → FTK Imager
   - Right-click → 'Run as Administrator' (required for raw disk access)
   - FTK Imager main window opens with empty Evidence Tree on left

2. Add Evidence Item
   - Menu: File → Add Evidence Item
   - Select Source: Physical Drive (for full forensic copy)
   - OR: Logical Drive (for single partition only)
   - Select drive: \\.\PHYSICALDRIVE0 (adjust for your system)
   - Click Next → Finish

3. Inspect the Evidence Tree
   The drive appears in the Evidence Tree panel (left side)
   Expand to see partitions, file systems, and directories
   NOTE: Do not open or modify any files — read-only mode only

4. Create Disk Image (E01 Format)
   - Right-click the drive → 'Create Disk Image...'
   - Select image type: E01 (EnCase) — recommended
   - Click 'Add...' to set image destination

5. Fill Evidence Information Form
   Evidence Number : EV-2026-001
   Unique Description : Server-Z Network Evidence 30-03-2026
   Examiner Name : SOC Analyst
   Notes : Source IP 10.0.2.20, INC-2026-0330-001

6. Configure Image Settings
   - Image Destination Folder : D:\Evidence\Images\
   - Image Filename : SERVER-Z_10.0.2.20_2026-03-30
   - Image Fragment Size : 0 (single file, no splitting)
   - Compression Level : 6 (balanced speed/size)
   - Verify images after they are created
   - Create directory listings of all files

7. Start Acquisition
   - Click 'Start' — imaging begins
   - Progress bar shows bytes read / estimated time remaining
   - DO NOT interrupt — any interruption invalidates the image
   - Typical time: ~30 min per 100 GB at USB 3.0 speed

8. Save Verification Report
   - After completion: File → Export Disk Image Information
   - Save as: D:\Evidence\SERVER-Z_Verification_2026-03-30.txt
   - This file contains MD5 + SHA1 + case metadata

2.5 Chain of Custody Documentation

Item #	Evidence Item	Description	Collected By	Date/Time	Hash (SHA256)	Storage Location
E00 1	Network Log Dump	Packet capture from Server-Z (10.0.2.20)	SOC Analyst	2026-04-02 14:30	a3f5...d9e2	Evidence Vault A
E00 2	Velociraptor Report	Process + netstat VQL output — JSON format	SOC Analyst	2026-04-02 14:45	b2c1...7f8a	Evidence Vault A
E00 3	Wazuh Alert Export	All alerts 30-03 to 03-04 — CSV export	SOC Lead	2026-04-02 15:00	c9d4...1b3e	Evidence Vault B
E00 4	Memory Dump	RAM image from compromised host (FTK Imager)	SOC Analyst	2026-04-02 15:30	d7e8...4a2f	Evidence Vault A

E00 5	Malwar e Sample	Downloaded .exe from 10.0.2.20 (quarantined)	AV System	2026-04-01 10:07	e1f2...8c5d	Malware Vault

Evidence Transfer Log:

Item ID	Transferred By	Received By	Date / Time	Transfer Method	Reason
E-0 01	SOC Analyst	Lead Investigator	2026-03-30 11:00 UTC	Secure USB (encrypted)	Network analysis review
E-0 02	SOC Analyst	Forensic Lab Tech	2026-03-30 11:30 UTC	Secure courier	Deep forensic examination
E-0 03	SOC Analyst	Forensic Lab Tech	2026-03-30 11:30 UTC	Secure courier	Memory forensics analysis
E-0 04	SOC Analyst	Incident Response Lead	2026-03-30 12:00 UTC	Encrypted email (TLS)	IOC extraction & reporting

Automated Chain-of-Custody Generation Script:

This PowerShell script automatically discovers all evidence files in a given directory, computes SHA256 + MD5
hashes for each, and generates a formal Chain-of-Custody report with timestamps. Run this script immediately
after evidence collection.

POWERSHELL — Generate-ChainOfCustody.ps1 (Full Script)

```
# ═══════════════════════════════════════════════════════════════════
# Script : Generate-ChainOfCustody.ps1
# Purpose : Auto-hash all evidence files and generate COC document
# Author : SOC Analyst
# Date : 30-03-2026
# Incident: INC-2026-0330-001 | Source: 10.0.2.20 (SERVER-Z)
# ═══════════════════════════════════════════════════════════════════
param(
[string]$EvidencePath = 'D:\Evidence',
[string]$Analyst = 'SOC Analyst',
[string]$IncidentID = 'INC-2026-0330-001',
[string]$SourceIP = '10.0.2.20',
[string]$Hostname = 'SERVER-Z'
)
$timestamp = Get-Date -Format 'yyyy-MM-dd HH:mm:ss UTC'
$reportPath = "$EvidencePath\ChainOfCustody_$($IncidentID)_$(Get-Date -Format 'yyyyMMdd').txt"
$report = @"
═══════════════════════════════════════════════════
CHAIN OF CUSTODY DOCUMENT
═══════════════════════════════════════════════════
Incident ID : $IncidentID
Source Host : $Hostname
Source IP : $SourceIP
Analyst : $Analyst
Generated On : $timestamp
Evidence Path : $EvidencePath
```

```
■■■
```

3 Flag connections on non-standard ports (not 80/443/53/22)
4 Correlate suspicious connections with process executable paths
5 Check executable paths — flag anything in Temp/AppData/Downloads
6 Search for C2 beacon indicators (regular interval connections)
7 Review PowerShell, WMI, and scheduled task activity

PHASE 4 DOCUMENTATION
1 Fill Chain-of-Custody table for all evidence items (EV-001 through EV-004)
2 Record all SHA256 hash values in COC document
3 Document every analyst action with timestamps
4 Run Generate-ChainOfCustody.ps1 to auto-generate hash report
5 Export full Velociraptor hunt results and artifact collections
6 Sign and date all evidence handling documentation

PHASE 5 REPORTING
1 Summarize all findings in Incident Analysis Report
2 List all suspicious connections with IPs, ports, and process names
3 Provide complete IOC list (IPs, domains, hashes, process names)
4 Include risk severity ratings for each finding
5 Recommend immediate containment actions
6 Recommend longer-term remediation steps

Master VQL Artifact — Complete Evidence Collection Package

VQL — Master Evidence Collection Artifact

```
-- ■■■
```

```
-- 6. Return all data with section labels
SELECT * FROM chain(
a={ SELECT 'NET_CONNECTIONS' AS Section, * FROM NetworkData },
b={ SELECT 'PROCESSES' AS Section, * FROM ProcessData },
c={ SELECT 'CORRELATION' AS Section, * FROM CorrelatedData },
d={ SELECT 'DNS_CACHE' AS Section, * FROM DNSData },
e={ SELECT 'LOGGED_USERS' AS Section, * FROM UserData }
)
```

SAMPLE INCIDENT FINDINGS REPORT

Suspicious Connections Identified

PROCESS	PID	REMOTE IP	PORT	STATE	SEVERITY	REASON
svchost.exe	2344	198.51.100.5	4444	ESTABLISHED	CRITICAL	System process connecting to external IP on Metasploit default port
powershell.exe	4812	203.0.113.10	8080	ESTABLISHED	HIGH	PowerShell making direct outbound connection — possible download cradle
unknown.exe	6621	192.0.2.88	1337	ESTABLISHED	CRITICAL	Unknown process name, non-standard path, leet-speak C2 port

Indicators of Compromise (IOCs)

IOC TYPE	VALUE	CONTEXT
IP Address	198.51.100.5	C2 Server — svchost.exe connection on port 4444
IP Address	203.0.113.10	C2/Download Server — powershell.exe on port 8080
IP Address	192.0.2.88	C2 Server — unknown.exe on port 1337
Process Name	unknown.exe	Unrecognized process with active C2 connection
Port	4444	Metasploit reverse shell default listener port
Port	1337	Common malware/C2 port (leet-speak)
Process Path	C:\Temp\unknown.exe	Executable running from %TEMP% — suspicious location

Recommendations

ACTION	PRIORITY	DETAILS

IMMEDIATE — Isolate	HIGH	Network-isolate SERVER-Z (10.0.2.20) from all internal and external networks immediately
IMMEDIATE — Block IPs	HIGH	Block outbound connections to 198.51.100.5, 203.0.113.10, 192.0.2.88 at perimeter firewall
IMMEDIATE — Kill Process	HIGH	Terminate unknown.exe (PID 6621) — document before killing
SHORT TERM — Sandbox	MEDIUM	Submit unknown.exe to malware sandbox (Hybrid Analysis/Any.run/VT)
SHORT TERM — Log Review	MEDIUM	Review PowerShell event logs (Event ID 4104) for script block content
SHORT TERM — Persistence	MEDIUM	Check scheduled tasks, registry run keys, and startup items for persistence
LONG TERM — Remediate	LOW	Rebuild SERVER-Z from clean image after evidence collection is complete
LONG TERM — Hunt	LOW	Run Velociraptor hunt across ALL endpoints for same IOCs

QUICK REFERENCE CHEAT SHEET

VQL QUERY	PURPOSE
SELECT * FROM netstat	All network connections (full dump)
SELECT * FROM netstat WHERE Status='ESTABLISHED'	Only active established connections
SELECT * FROM pslist()	All running processes with full details
SELECT * FROM dns_cache()	Recent DNS lookups (cached resolutions)
SELECT * FROM logged_in_users()	Currently and recently logged-in users
SELECT * FROM users()	All user accounts on the system
SELECT * FROM services()	All Windows services and their status
SELECT * FROM interfaces()	Network interface configuration
SELECT * FROM arp()	ARP cache — local network neighbors
SELECT * FROM environ()	Environment variables for all processes

VELOCIRAPTOR ARTIFACT	PURPOSE

Windows.Forensics.Prefetch	Prefetch files — program execution history
Windows.EventLogs.Evtx	Parse Windows Event Log (.evtx) files
Windows.Persistence.PersistenceClientMon	Monitor persistence locations (registry, startup)
Windows.System.ScheduledTasks	List all scheduled tasks (possible persistence)
Windows.Sys.StartupItems	Startup programs — Run keys and startup folder
Windows.Memory.VAD	Virtual Address Descriptor — memory segments
Windows.Network.InterfaceAddresses	All IP addresses and network interfaces

FTK Imager CLI Command Reference

CMD — FTK Imager CLI Commands
```
#  FTK Imager CLI — Common Commands
# Location: C:\Program Files\AccessData\FTK Imager\ftkimager.exe
# Create E01 image (recommended — compressed + verified)
ftkimager.exe \\.\PHYSICALDRIVE0 D:\Evidence\SERVER-Z --e01 --verify --compress 6
# Create raw DD image (no compression, maximum compatibility)
ftkimager.exe \\.\PHYSICALDRIVE0 D:\Evidence\SERVER-Z --dd
# Create image of a specific logical drive (partition)
ftkimager.exe \\.\C: D:\Evidence\SERVER-Z_C_Drive --e01 --verify
# Capture memory (live RAM)
ftkimager.exe --memory D:\Evidence\SERVER-Z_Memory
# Verify an existing E01 image
ftkimager.exe D:\Evidence\SERVER-Z.E01 --verify
# List files in an E01 image
ftkimager.exe D:\Evidence\SERVER-Z.E01 --list-files > D:\Evidence\file_listing.txt
```
Evidence Hashing Quick Reference

POWERSHELL + BASH — Evidence Hashing Commands
```
#  Windows PowerShell
# SHA256 of a single file
Get-FileHash -Path 'D:\Evidence\file.E01' -Algorithm SHA256
# MD5 of a single file
Get-FileHash -Path 'D:\Evidence\file.E01' -Algorithm MD5
# Hash ALL files in Evidence directory (recursive)
Get-ChildItem 'D:\Evidence' -Recurse -File |
ForEach-Object { Get-FileHash $_.FullName -Algorithm SHA256 } |
Export-Csv 'D:\Evidence\all_hashes.csv' -NoTypeInformation
#  Linux Bash
# SHA256
sha256sum /evidence/SERVER-Z_2026-03-30.E01
# MD5
md5sum /evidence/SERVER-Z_2026-03-30.E01
# Hash all files in a directory
find /evidence -type f -exec sha256sum {} \; > /evidence/hashes.txt
```

Common Ports — Legitimate vs Suspicious

PORT	SERVICE	STATUS	NOTES
80	HTTP	Legitimate	Standard web traffic
443	HTTPS	Legitimate	Encrypted web — but also Cobalt Strike C2
53	DNS	Legitimate	Domain resolution — watch for DNS tunneling
22	SSH	Legitimate	Secure shell — verify authorized use
3389	RDP	Monitor	Remote Desktop — common brute-force target
445	SMB	Monitor	File sharing — WannaCry/NotPetya vector
4444	Metasploit	SUSPICIOUS	Default Meterpreter reverse shell
4443	Cobalt Strike	SUSPICIOUS	Common Cobalt Strike HTTPS listener
31337	Elite / Back Orifice	SUSPICIOUS	Historical trojan — still used
1337	Custom RAT	SUSPICIOUS	Leet-speak — custom malware C2
6666	IRC Bot	SUSPICIOUS	IRC-based botnet C2 channels
9999	RAT	SUSPICIOUS	Common remote access trojan port